

My Name

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EDUCATION

National University of Computer And Emerging Sciences, Chiniot Faisalabad Campus

Bachelor of Science, Computer Science - 2026

(3.59 / 4)

Nixor College Karachi

Alevels - Computer Science - 2022

4A*

PROJECTS

SoccerVision

- Developed an end-to-end AI system for automated football match highlight generation
- Implemented computer vision models to detect key events (goals, penalties, freekicks etc.)
- Generated multiple highlight types (full match, event-based, match summaries) from raw footage
- Integrated APIs for payments, authentication and social media sharing.

QLoRA Fine-Tuning of Vision Language Models

- Fine-tuned the vision-language model Qwen Qwen2-VL-2B-Instruct using QLoRA for document image to Markdown conversion.
- Applied parameter-efficient fine-tuning techniques including 4-bit quantization, LoRA adapters, gradient accumulation, and GPU-based training on Kaggle.

Masked Auto Encoder (MAE)

- Built a self-supervised MAE using ViT (Base + Small) for image reconstruction
- Applied 75% patch masking and asymmetric encoder-decoder design
- Optimized training with mixed precision on dual T4 GPUs (Kaggle)

Custom Diffusion Model for Image Generation

- Implemented a Denoising Diffusion Probabilistic Model (DDPM) in PyTorch using a custom U-Net architecture for high-resolution image generation and reconstruction from noisy inputs.
- Developed the complete forward and reverse diffusion pipeline, including noise scheduling, timestep embeddings, mixed-precision training, and image quality evaluation using PSNR and SSIM metrics

Image Captioning System

- Built a multimodal image captioning model using ResNet50 + LSTM (Seq2Seq)
- Extracted and cached 2048-d image features to optimize training

Traffic Sign Detection & Classification

- Built a computer vision pipeline using YOLO and CNN models for detection and classification tasks
- Fine Tuned models on traffic signs dataset

Activities, Honour and Awards

6x Deans List, FAST NUCES Chiniot-Faisalabad

AI & Computer Network TA, FAST NUCES Chiniot-Faisalabad

99.3 Percentile in NSCT TEST, NUCES, HEC

3rd Position FPSC coding competition, FAST NUCES Chiniot-Faisalabad

TECHNICAL EXPERTISE

Languages: Python, C++, SQL

Frameworks & Libraries: PyTorch, NumPy, FastAPI, Scikit-learn, LangChain

Tools & Platforms: Git, Docker, AWS, Ollama, Kaggle, Firebase

Interests & Hobbies

GeoPolitics, History, Sports